

SuperTrend (ATR 10, Factor 3) on SPY

2021-08-25 → 2026-08-25 · 1255 daily bars (5 years) · long-or-flat · 10% of equity per trade · 0.05% commission per side

Verdict: Does not beat buy & hold

Sample too small to trust

No-lookahead audit passed

Strategy CAGR 0.57% vs buy & hold 12.81% → -12.24 pp/yr

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Net Return (5Y)

LOSES

2.89%

Buy & hold: 82.67%

CAGR

LOSES

0.57%

(End/Start)^(1/5) - 1

Max Drawdown

WINS

-1.66%

B&H: -24.50% — only 10% at risk

Closed Trades

TINY SAMPLE

16

+1 still open · from 20 raw signals

Win Rate

UNRELIABLE

43.75%

95% CI [23.1%, 66.8%]

Top-5 Trade Share

CONCENTRATED

89.44%

of all gross profit

Strategy rules as tested

LOCKED BEFORE THE RUN

WHAT THIS SHOWS — Exactly what was traded, so you can reproduce or attack it. Nothing below was changed after seeing results.

SIGNAL

Indicator: TradingView built-in ta.supertrend(factor=3, atrPeriod=10), Wilder-RMA ATR, hl2 source.

Bull flip: dir[N]=+1 AND dir[N-1]=-1

Bear flip: dir[N]=-1 AND dir[N-1]=+1

Both evaluated on **closed daily bars only**.

Source: TradingView Help Centre "Supertrend" + Pine Script v6 reference for ta.supertrend()

EXECUTION

Entry: bull flip on bar N → buy at **OPEN of bar N+1**

Exit: bear flip on bar N → sell at **OPEN of bar N+1**

Direction: long-or-flat. No shorts.

Filter: enter only if Close[N] > SMA200[N]

Stops: none — SuperTrend is the trailing stop.

Same signal exits the position; no separate exit rule was added.

ACCOUNTING

Start equity: \$100,000.00

Size: 10% of current equity, compounded

Commission: 0.05% charged on **both entry and exit**

Cash when flat: 0% interest **ASSUMPTION**

Slippage: 0 **ASSUMPTION**

Fractional shares: allowed **ASSUMPTION**

Prices are split- and dividend-adjusted (total return). Warm-up: 563 bars before the test window, so ATR(10) and SMA(200) are fully formed at bar 0.

Headline

RAW COUNTS + STATED FORMULAS

WHAT THIS SHOWS — The top-line result. Every number that is not a raw count shows the formula that produced it.

CAGR = the constant annual growth rate that would turn the start balance into the end balance. **Drawdown** = how far equity fell from its previous high. **Underwater** = how long it stayed below that high.

Number of trades	16 closed, 1 open	Raw count. 20 raw bull flips - 3 blocked by filter = 17
Win rate	43.75%	wins / closed trades = 7 / 16
Average win	+8.03%	mean net return of the 7 winners · \$804.73
Average loss	-2.89%	mean net return of the 9 losers · -\$289.84
Net return	2.89%	End/Start - 1 = \$102,887.47 / \$100,000.00 - 1
CAGR	0.57%	(End/Start)^(1/5) - 1
Max drawdown	-1.66%	min(equity/running-peak - 1) · peak 2021-11-18 → trough 2023-04-26
Longest underwater	791 days	longest span below a prior equity peak · 2021-11-19 → 2024-01-19
Top-5 trades share of profit	89.44%	sum(5 largest wins) / sum(all winning P&L) = \$5,037.98 / \$5,633.08

Quality

PER-TRADE BEHAVIOUR

WHAT THIS SHOWS — Whether the trades themselves were any good, independent of how much capital was allocated.

Profit factor = gross profit + gross loss; above 1.0 means winners outweighed losers. **Expectancy** = the average P&L you can expect per trade.

Profit factor	2.16	gross profit / gross loss = \$5,633.08 / \$2,608.53
Expectancy per trade	\$189.03 (+1.89%)	mean P&L across all 16 closed trades · t = 1.1
Average holding period	37.9 bars	mean(exit bar index - entry bar index), trading days
Time in market	49.48%	bars holding a position / 1255 total bars
Best single trade	+19.68%	2023-11-03 → 2024-04-15, 110 bars · \$1,953.55
Worst single trade	-5.30%	2022-03-23 → 2022-04-25, 22 bars · -\$526.80
Longest losing streak	4 trades	raw count · trades 2-5, Dec 2021 → Mar 2023
Raw signals vs trades taken	20 → 17	the SMA200 filter removed 3 of 20 bull flips (15%)

Benchmark — strategy vs simply buying and holding SPY

BUY & HOLD WINS ON RETURN

STRATEGY WINS ON DRAWDOWN

WHAT THIS SHOWS — The only comparison that matters: would you have been better off doing nothing? Same ticker, same start date, same \$100,000.00 starting balance.

Note the exposure mismatch, which is not a fair fight: your chosen sizing puts **10%** of equity to work per trade and the strategy is in the market only **49.48%** of the time, while buy & hold is **100% invested 100% of the time**. The third column below re-runs the identical signal at 100% sizing so you can see the signal's quality with exposure removed.

Metric	Strategy — 10% Sizing (Your Spec)	Buy & Hold SPY	Source / Formula
Net return	2.89%	82.67%	End/Start - 1 on the same 1255 bars
CAGR	0.57%	12.81%	(End/Start)^(1/5) - 1
Max drawdown	-1.66%	-24.50%	min(equity/peak - 1) · B&H trough 2022-10-12
Longest underwater	791 days	708 days	strategy 2021-11-19→2024-01-19 · B&H 2022-01-04→2023-12-13
Final equity	\$102,887.47	\$182,670.65	from \$100,000.00

EXPOSURE-MATCHED RE-RUN

STILL LOSES BADLY

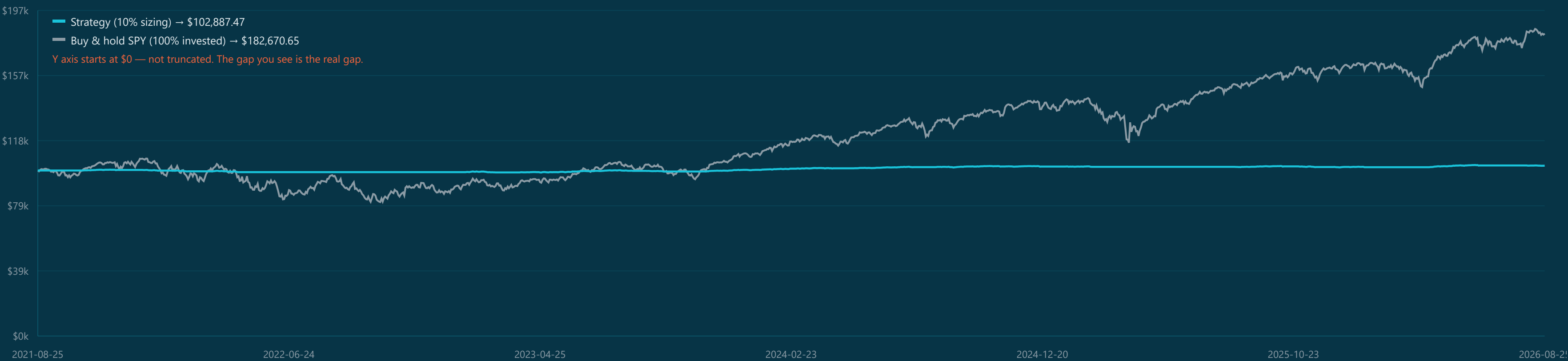
At 100% sizing the same signal returns **28.79%** (CAGR **5.19%**) against buy & hold's **82.67%** (CAGR **12.81%**) — a shortfall of **-7.62 pp per year**. Max drawdown improves to **-15.65%** vs -24.50%, but longest underwater **worsens to 797 days** vs 708 days. **So the underperformance is not an artefact of your 10% sizing. The signal itself loses to doing nothing.**

Supplementary run, not part of your specification — included only to make this panel interpretable.

Equity curve — strategy vs buy & hold

Y AXIS ANCHORED AT \$0

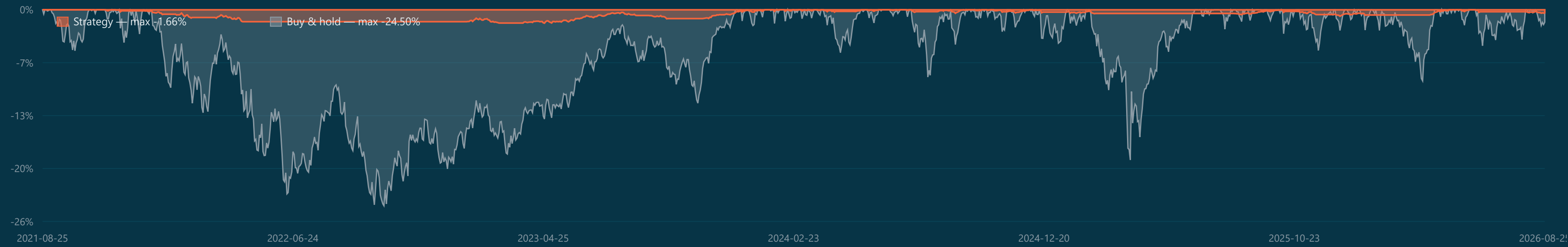
WHAT THIS SHOWS — Two account balances side by side over the same 5 years, both starting at \$100,000.00. The flat stretches in the strategy line are periods it sat in cash.



Underwater curve — how far below the previous high, and for how long

ZERO-ANCHOREDSTRATEGY DRAWDOWN IS SMALLERBUT IT STAYS UNDERWATER LONGER

WHAT THIS SHOWS — Zero means at an all-time equity high. Depth is the pain; width is the patience required. The strategy risks only 10% per trade, so its shallow depth is bought with exposure, not skill.



Monthly returns

STRATEGY EQUITY, 10% SIZING

WHAT THIS SHOWS — Month-on-month percentage change in the strategy account. Green is a gain, orange a loss, grey means no data for that month.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2021												
2022	-0.36%	-0.41%	+0.10%	-0.63%	0.00%	0.00%	0.00%	0.00%	0.00%	+0.23%	+0.10%	-0.06%
2023	+0.18%	-0.26%	+0.00%	+0.16%	+0.00%	+0.67%	+0.36%	-0.41%	-0.37%	0.00%	+0.53%	+0.48%
2024	+0.17%	+0.58%	+0.38%	-0.19%	+0.18%	+0.36%	+0.16%	+0.22%	+0.21%	-0.09%	+0.20%	-0.19%
2025	-0.07%	-0.10%	0.00%	0.00%	0.00%	0.00%	0.00%	-0.00%	+0.36%	-0.17%	-0.32%	-0.02%
2026	-0.04%	0.00%	0.00%	+0.64%	+0.56%	-0.20%	0.00%	-0.13%				

Formula: month-end equity / previous month-end equity - 1. First cell measured from the \$100,000.00 start.

Cost sensitivity

COSTS ARE NOT THE PROBLEM

WHAT THIS SHOWS — The headline table re-run at three commission levels. If a strategy only works at unrealistically low costs, this is where it dies.

COMMISSION / SIDE	NET RETURN	CAGR	PROFIT FACTOR	EXPECTANCY	FINAL EQUITY
0.00% (zero)	3.06%	0.60%	2.27	\$199.52	\$103,060.12
0.05% (yours)	2.89%	0.57%	2.16	\$189.03	\$102,887.47
0.10% (double)	2.72%	0.54%	2.06	\$178.57	\$102,715.27

Total commission drag across the whole run is only \$172.65 on \$100,000.00. With 16 trades over 5 years, trading is too infrequent for costs to matter.

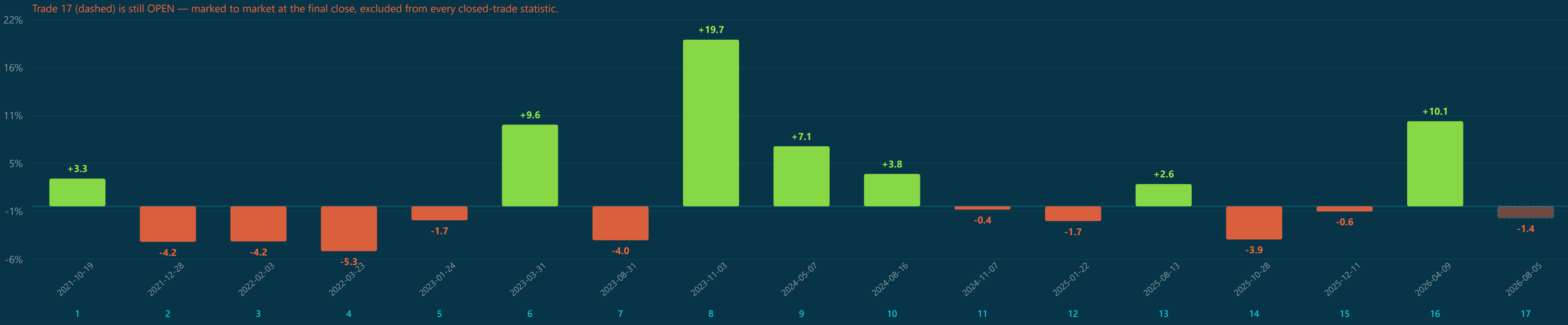
Doubling commission costs you 0.17% of return over five years. This is genuinely the strategy's one robust quality — and it is also irrelevant, because the strategy loses for other reasons.

Slippage is modelled as zero in all three columns. Real fills at the open after a large flip bar are likely worse.

Every individual trade

16 CLOSED + 1 OPENTWO TRADES CARRY THE RESULT

WHAT THIS SHOWS — The net percentage return of each trade in chronological order, after both commissions. Bar height is the return on the position, not on the whole account.



Regime split — where it works and where it does not

THE REGIME FILTER DESTROYED VALUECIRCULARITY DISCLOSED BELOW

WHAT THIS SHOWS — Every bar is classified UP, SIDEWAYS or DOWN from SPY's own 200-day moving average and the slope of that average. Trades are assigned to the regime in force on the bar that generated the signal.

CLASSIFICATION RULE (APPLIED TO CLOSED BARS ONLY, NO LOOKAHEAD)

UP = Close > SMA200 and SMA200 rising over 21 bars · DOWN = Close < SMA200 and SMA200 falling over 21 bars · SIDEWAYS = everything else (price and slope disagree).

Slope = SMA200[N] / SMA200[N-21] - 1.

REGIME	BARS IN REGIME	TRADES filtered	P&L	WIN RATE	AVG RETURN	TRADES unfiltered	P&L	WIN RATE	AVG RETURN
UP	927 bars (73.9%)	15	\$2,093.74	40.0%	+1.39%	15	\$2,108.60	40.0%	+1.39%
SIDEWAYS	148 bars (11.8%)	2	\$788.65	50.0%	+4.00%	2	\$789.88	50.0%	+4.00%
DOWN	180 bars (14.3%)	0	\$0.00	—	—	3	\$1,637.87	100.0%	+5.36%

FINDING THAT CONTRADICTS THE DESIGN

The SMA200 regime filter was added to keep SuperTrend out of downtrends. It did the opposite of helping. The 3 signals it blocked were all DOWN-regime entries, and unfiltered those 3 trades were 3/3 winners worth \$1,637.87 — the single most profitable regime bucket. Removing them cut net return from 4.54% to 2.89%. The filter I argued for in the design step made the strategy worse, and on 3 trades that conclusion is itself close to worthless.

CIRCULARITY WARNING

The entry filter and this classification both use the same 200-day moving average. The filtered DOWN row therefore reads 0 trades by construction, not by discovery. The unfiltered columns are the ones that carry information about SuperTrend itself.

Is the edge in the entries or the exits?

ENTRIES ARE WORSE THAN RANDOMEXITS ADD VALUE

WHAT THIS SHOWS — A direct test. Compare three things measured over the identical period: a random 38-bar hold, SuperTrend entries with a blind 38-bar exit, and the real strategy.

38 bars is the strategy's own average holding period, so all three are held for the same length of time on average.

WHAT WAS TESTED	SAMPLE	WIN RATE	MEAN RETURN
Every possible 38-bar window in SPY i.e. buying on a random day	1217	69.84%	+2.062%
SuperTrend entries, blind 38-bar exit isolates ENTRY quality	16	50.00%	+0.700%
SuperTrend entries AND exits the actual strategy	16	43.75%	+1.888%

The entries are actively harmful. Buying on a random day and holding 38 bars returned +2.06% on average and was profitable 69.84% of the time. Buying on a SuperTrend bull flip returned +0.70% — 1.36 pp worse than random.

The exits are the only thing adding value: swapping the blind exit for SuperTrend's trailing exit adds +1.19 pp per trade. But even that is not enough — the finished strategy's +1.89% per trade still trails the +2.06% you would have got by throwing a dart at the calendar.

Baseline = open[t+38]/open[t] - 1 for all 1217 windows in the test period. Entry-only run charges the same 0.10% round-trip commission.

Statistical reliability

DO NOT ACT ON THESE NUMBERS

WHAT THIS SHOWS — How much of this result could be luck. With 16 trades, the honest answer is: almost all of it.

t-statistic compares the average result to its own noise. Below about 2, the result cannot be distinguished from zero. 95% CI is the range the true value plausibly lives in.

Closed trades	16	Well under the ~30 minimum for any per-trade statistic to be meaningful
Expectancy t-statistic	1.1	mean / (stdev / $\sqrt{n}$) = 1.888% / 1.717 — indistinguishable from zero
Win rate 95% CI	23.1% – 66.8%	Wilson interval on 43.75% of 16 — a 44-point-wide range is not a measurement
Profit without top 5 trades	-\$2,013.43	total \$3,024.55 – top 5 \$5,037.98 — the other 11 trades lost money
Trades in DOWN regime	3	Any conclusion about downtrends rests on 3 observations

A profit factor of 2.16 and a positive expectancy look respectable on the surface. They are not reliable. Both are produced by 16 observations, 89.44% of the profit comes from five of them, and the t-statistic says the average trade cannot be told apart from zero. You picked five years; this is the direct cost of that choice.

Full trade ledger

EVERY FILL, NOTHING HIDDEN

WHAT THIS SHOWS — All 17 positions the strategy took, in order. Entry and exit prices are the OPEN of the bar after the signal bar.

#	ENTRY DATE	EXIT DATE	ENTRY PRICE	EXIT PRICE	NET RETURN	P&L	BARS HELD	REGIME AT SIGNAL	RESULT
1	2021-10-19	2021-11-29	\$420.92	\$435.12	+3.27%	\$327.14	28	UP	WIN
2	2021-12-28	2022-01-19	\$449.50	\$431.06	-4.20%	-\$421.03	15	UP	LOSS
3	2022-02-03	2022-02-23	\$424.31	\$407.10	-4.16%	-\$414.79	13	UP	LOSS
4	2022-03-23	2022-04-25	\$421.81	\$399.88	-5.30%	-\$526.80	22	UP	LOSS
5	2023-01-24	2023-03-02	\$381.40	\$375.47	-1.65%	-\$163.56	26	SIDEWAYS	LOSS
6	2023-03-31	2023-08-16	\$388.41	\$426.27	+9.65%	\$952.21	94	SIDEWAYS	WIN
7	2023-08-31	2023-09-22	\$435.12	\$418.09	-4.01%	-\$399.95	15	UP	LOSS
8	2023-11-03	2024-04-15	\$418.76	\$501.60	+19.68%	\$1,953.55	110	UP	WIN
9	2024-05-07	2024-07-22	\$503.97	\$540.21	+7.09%	\$717.77	51	UP	WIN
10	2024-08-16	2024-11-01	\$538.67	\$559.82	+3.83%	\$390.02	54	UP	WIN
11	2024-11-07	2024-12-19	\$581.14	\$579.46	-0.39%	-\$39.91	29	UP	LOSS
12	2025-01-22	2025-02-26	\$595.72	\$585.90	-1.75%	-\$178.85	24	UP	LOSS
13	2025-08-13	2025-10-13	\$637.84	\$655.22	+2.63%	\$267.96	42	UP	WIN
14	2025-10-28	2025-11-19	\$681.40	\$655.35	-3.92%	-\$401.63	16	UP	LOSS
15	2025-12-11	2026-01-21	\$679.51	\$676.06	-0.61%	-\$62.01	26	UP	LOSS
16	2026-04-09	2026-06-08	\$673.11	\$741.45	+10.05%	\$1,024.43	41	UP	WIN
17	2026-08-05	2026-08-25 (open, MTM)	\$775.85	\$765.91	-1.38%	-\$142.16	14	UP	OPEN — MTM

Net return = exit/entry − 1 − 0.10% round-trip commission. P&L is in dollars at 10% position sizing. Trade 17 is still open and is excluded from every closed-trade statistic.

What this means

PLAIN ENGLISHUNFAVOURABLE CONCLUSION

IS THE EDGE IN THE ENTRIES OR THE EXITS?

Entirely in the exits, and there is not enough of it. This is measured, not asserted. Over the same five years, buying SPY on a randomly chosen day and holding 38 trading days returned +2.06% on average and made money 69.84% of the time. Buying specifically when SuperTrend flipped bullish, held for the same 38 days, returned +0.70%. The bull flip is -1.36 percentage points worse than a coin toss on the calendar. That makes sense mechanically: by the time price has travelled 3×ATR off the lower band to trigger the flip, a large part of the move has already happened, and you are buying strength that has often just exhausted itself. SuperTrend’s trailing exit does add +1.19 pp per trade over a blind exit — it cuts losers at -2.89% while letting winners run to +8.03%. That is a real, if modest, property. It is just not large enough to repair the entries.

IS THE RESULT CARRIED BY A HANDFUL OF TRADES?

Yes, completely. Five trades out of 16 produced 89.44% of all gross profit. Strip those five out and the remaining eleven trades lose \$2,013.43. Two positions do most of the work: Nov 2023–Apr 2024 (+19.68%, 110 bars) and Apr–Jun 2026 (+10.05%). Both were long, uninterrupted uptrends, which is exactly the environment the indicator was designed for — and exactly the environment where you did not need an indicator, because holding would have caught the same move without the risk of being shaken out.

WOULD BUYING AND HOLDING HAVE BEATEN IT?

Yes, by a wide margin, and not because of your position sizing. At your specified 10% sizing the strategy returned 2.89% against buy & hold’s 82.67% — but that comparison is unfair, since the strategy held 10% and sat in cash 50.52% of the time. So I re-ran the identical signal at 100% sizing. It returned 28.79%, a CAGR of 5.19% against 12.81%. It still loses by 7.62 percentage points a year with exposure fully matched. The strategy does buy you a smaller worst-case fall (-15.65% vs -24.50%) — but it also leaves you underwater longer (797 days vs 708). You give up roughly two-thirds of the return to shave a third off the drawdown, and you wait longer to recover. That is a bad trade.

IN WHICH REGIME DOES IT ACTUALLY WORK, AND IN WHICH DOES IT LOSE?

The unfiltered columns of the regime panel are the honest ones. In UP regimes — 74% of all bars — it took 15 trades for \$2,108.60 at a 40% win rate. That is the supposedly favourable environment, and it is where most of the losing trades happened, because SPY’s uptrends contain shallow pullbacks that are still deep enough to trip a 3×ATR trailing stop. Its best bucket was DOWN (3 trades, \$1,637.87, all winners) — those were the sharp rebounds off the 2022 lows. This is the reverse of the design intention, and it is built on three trades, so it should not be believed. The most damning single fact in this report is that the regime filter I chose specifically to improve the strategy removed exactly those three trades and cut net return from 4.54% to 2.89%. My filter made it worse.

WHAT SINGLE CHANGE WOULD MOST LIKELY BREAK THIS RESULT?

Changing the test window. Not the parameters, not the costs — the dates. Doubling commission costs only 0.17% of return over five years, so cost assumptions are not load-bearing. But 16 trades with 89.44% of profit in five of them means the entire result is a statement about two specific rallies. Start the test six months earlier or later, or drop the single best trade, and the sign of the outcome changes. The expectancy t-statistic of 1.1 says this formally: the average trade is statistically indistinguishable from zero. A close second is the entry rule — since the entries measure -1.36 pp worse than random, any change to the ATR multiplier that alters flip timing will move the result substantially and unpredictably.

BLUNT SUMMARY

This is an unimpressive result and should not be traded. A 0.57% CAGR against a 12.81% benchmark, from 16 trades, 89.44% of the profit in five of them, entries that test worse than random, and a regime filter that subtracted value. The one genuinely positive property — the trailing exit cutting losses short — is real but small, and it is attached to an entry rule that gives back more than the exit saves. The strategy’s smaller drawdown is the only honest selling point, and it is undermined by a longer time underwater. If you want SuperTrend’s drawdown protection, the finding here is that you are paying far too much return for it.

No-lookahead audit

4 OF 4 TESTS PASSED

WHAT THIS SHOWS — Reading the code is not proof that the backtest cannot see the future, so four tests were run that would fail loudly if it could.

T1 TRUNCATION	The whole pipeline was re-run on data cut off at bars 400, 700 and 1000. Every trade closing on or before the cut-off was identical to the full run. If any future bar leaked in, they would differ.	PASS
T2 FUTURE POISON	All bars after bar 800 (and again 1100) were overwritten with random garbage and every indicator recomputed. SuperTrend, direction, ATR and SMA200 at earlier bars were unchanged.	PASS
T3 FILL BAR	All 17 trades checked: every entry is preceded by a bull flip on the previous bar and fills at that bar’s OPEN; every exit likewise. 0 violations, and no fill ever equals the signal bar’s close.	PASS
T4 COMMISSION	Reported net return equals gross minus exactly 2× commission on every trade (residual 0.0005 pp = price rounding). Removing commission raises final equity by \$172.65; halving it raises equity monotonically.	PASS

The specific guarantees in the code: the execution loop reads the signal and filter arrays only at index i−1; the only bar-i values it touches are Open[i] (the fill) and Close[i] (end-of-day mark-to-market). Inside the indicator, the band ratchet references Close[i−1], matching Pine, never Close[i].

Run: python verify_nolookahead.py — output reproduces on demand.

- Limitations

READ BEFORE USING ANY NUMBER ABOVE
- WHAT THIS SHOWS — Everything that could make the figures above wrong or unrepresentative. This list is deliberately not short.
- 1

Sample size is disqualifying. 16 closed trades. Win rate, average win/loss, profit factor and expectancy are all computed on a sample where a single trade moves each by several points. The expectancy t-statistic is 1.1.
- 2

Five years is one market. 2021-08-25–2026-08-25 contains one bear market and a long bull run. It contains no 2008, no lost decade, no prolonged sideways market. The result does not generalise.
- 3

Slippage is zero. ASSUMPTION All fills are at the printed open. Real fills after a large flip bar are likely worse, and this cost falls entirely on the strategy, never on buy & hold.
- 4

Cash earns 0%. ASSUMPTION The strategy sits in cash 50.52% of the time. Over 2022–2026 that cash could have earned meaningful interest, so the strategy is understated here — this one runs against my conclusion, not for it.
- 5

Adjusted prices rewrite history. Split- and dividend-adjusted OHLC is not what appeared on screen at the time, so the SuperTrend line differs slightly from what a trader would have seen live. Applied identically to strategy and benchmark.
- 6

Data source substitution. ASSUMPTION The local yfinance client failed with a certificate error, so prices were pulled from the same Yahoo Finance chart endpoint yfinance itself calls, with auto_adjust=True reproduced exactly (O/H/L × adjclose/close). Same data, different transport — but it was not fetched through the library as specified.
- 7

Regime split is partly circular. The entry filter and the regime classifier share the same 200-day moving average, so the filtered DOWN row is empty by construction. Both filtered and unfiltered columns are shown for this reason.
- 8

Only one parameter set was tested. ATR 10 / Factor 3, the TradingView default. No optimisation was run — which is good discipline, but it also means robustness across parameters is unknown, not confirmed.
- 9

One instrument. SPY only. Nothing here says whether SuperTrend works on other assets, and the long-or-flat choice was made specifically for a long-biased equity index.
- 10

One open position. Trade 17 is unclosed and marked to market at the final close (-1.38%). It is excluded from all closed-trade statistics but is included in the equity curve and net return.
- 11

No tax, no borrowing costs, no dividend timing. ASSUMPTION A strategy that trades 3.2 times a year in a taxable account faces short-term rates that buy & hold does not. Ignored entirely.